

A STABLE MULTISTEP SCHEME FOR SOLVING BACKWARD STOCHASTIC DIFFERENTIAL EQUATIONS*

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Abstract. In this paper we propose a stable multistep scheme on time-space grids for solving backward stochastic differential equations. In our scheme, the integrands, which are conditional mathematical expectations derived from the original equations, are approximated by using Lagrange interpolating polynomials with values of the integrands at multiple time levels. They are then numerically evaluated using the Gauss–Hermite quadrature rules and polynomial interpolations on the spatial grids. Error estimates are rigorously proved for the semidiscrete version of the proposed scheme for backward stochastic differential equations with certain types of simplified generator functions. Finally, various numerical examples and comparisons with some other methods are presented to demonstrate high accuracy of the proposed multistep scheme.

Key words. backward stochastic differential equation, multistep scheme, Gauss–Hermite quadrature rule, time-space grid

AMS subject classifications. 60H35, 60H10

DOI. 10.1137/09076979X

1. Introduction. In this paper, we consider numerical solution of the following backward stochastic differential equation (BSDE):

$$(1.1) \quad \begin{cases} -dy_t = f(t, y_t, z_t)dt - z_t dW_t, & t \in [0, T], \\ y_T = \xi, \end{cases}$$

where T is a fixed positive number, W_t is the standard d -dimensional Brownian motion defined on a complete filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_{0 \leq t \leq T})$, $f(t, y, z)$ is an adapted stochastic process with respect to $\{\mathcal{F}_t\}$ ($0 \leq t \leq T$) for each (y, z) , and ξ is an $\{\mathcal{F}_T\}$ -measurable random variable. Existence and uniqueness of the solution of the BSDE (1.1) were proved in [25] by Pardoux and Peng in 1990. Since then, BSDEs and their solutions have been extensively studied by many researchers. In [27], Peng obtained a direct relation between forward-backward stochastic differential equations (FBSDEs) and PDEs, and then in [26], he also found the maximum principle for stochastic control problems. Many important properties of BSDEs and their applications in finance were studied by El Karoui, Peng, and Quenez in [15]. An important concept of g -expectation was introduced in [28] for the solution of BSDEs,

*Received by the editors September 2, 2009; accepted for publication (in revised form) June 17, 2010; published electronically August 24, 2010. The U.S. Government retains a nonexclusive, royalty-free license to publish or reproduce the published form of this contribution, or allow others to do so, for U.S. Government purposes. Copyright is owned by SIAM to the extent not limited by these rights.

<http://www.siam.org/journals/sinum/48-4/76979.html>

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and subsequently it was also found in [11] that a dynamic coherent risk measure can be valued by a properly defined g -expectation.

While analytic solutions of BSDEs are often very difficult to obtain, approximate numerical solutions of BSDEs are relatively easier to compute and thus become highly desired in practical applications. There are two main families of numerical methods for solving BSDEs. One is proposed based on the relation between the BSDE and its corresponding parabolic PDE. A typical four-step method for studying FBSDEs (not a numerical method) was first developed in [18] using this relation. Based on similar ideas derived in the four-step method, some algorithms were proposed to numerically solve FBSDEs; see [10, 19, 20, 21, 23, 24, 32] for details. When the corresponding parabolic PDE is quasi-linear, Delarue and Menozzi proposed in [8] a special time-space forward-backward discretization scheme based on the theory of fully coupled FBSDEs, and later in [9] they improved the proposed scheme through an interpolation procedure.

The other approach is developed directly based on the BSDE [2, 3, 4, 6, 7, 8, 12, 17, 22, 29, 31, 33, 34]. In [6], Chevance suggested an effective time-space discretization scheme when the function f does not depend on the variable z_t . Bally proposed in [2] a numerical discretization when f depends on the variables y_t and z_t ; in particular, the time discretization is performed on a random net to overcome the difficulties of discretizing the stochastic integral. In [3], Bender and Denk introduced a forward scheme for solving BSDEs. In [29], Peng proposed an iterative linear approximation algorithm which converges under reasonable assumptions. Some random walk methods for solving BSDEs were suggested by Mémin, Peng, and Xu in [22], and convergence of those methods was also proven there. Under certain weaker regularity assumptions, Zhang developed a modified numerical scheme for solving BSDEs and studied its convergence rate in [31], and then Cvitanic and Zhang proposed some Monte Carlo methods for solving FBSDEs in [7]. In [12], Gobet and Labart generalized Zhang's result and obtained error estimates of the scheme measured in the strong L^p -sense ($p \geq 1$). Bouchard and Touzi also presented an implicit scheme in [4] based on Zhang's work. Zhao, Chen, and Peng proposed a numerical method with high accuracy for solving BSDEs in [33] and Zhao, Wang, and Peng studied its error estimates in [34]. We note that all numerical methods reviewed above for solving BSDEs or FBSDEs are one-step schemes in the time direction.

In this paper, we propose a stable multistep scheme on the time-space grids for numerically solving BSDEs. In the proposed scheme, the integrands, which are conditional mathematical expectations derived from the original equations, are approximated by using the Lagrange interpolating polynomials with values of the integrands at multiple time levels. In particular, they are then numerically evaluated using the Gauss–Hermite quadrature rules and polynomial interpolations on the spatial grids. When the generator f is independent of z_t , we rigorously prove that the convergence of the proposed semidiscrete scheme can be of high order, and the exact order depends on the number of time levels used.

The rest of the paper is organized as follows. Some preliminaries are first given in section 1.1. Reference equations for the multistep scheme are derived from the BSDE (1.1) in section 2, and in section 3 we then present a stable multistep scheme for their numerical discretizations. Error estimates of the proposed semidiscrete scheme are rigorously proven while the generator function f is independent of z_t in section 4. Extensive numerical tests and comparisons are given in section 5, and these results are shown to be consistent with the theoretical ones. Finally some concluding remarks are given in section 6.

1.1. Preliminaries. Let $\{\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_{0 \leq t \leq T}\}$ be a complete, filtered probability space on which a standard d -dimensional Brownian motion W_t is defined, such that $\{\mathcal{F}_t\}_{0 \leq t \leq T}$ is the natural filtration of the Brownian motion W_t and all the $\mathbb{P}$ -null sets are augmented to each σ -field $\mathcal{F}_t$. Define $|\cdot|$ and $L^2 = L^2_{\mathcal{F}}(0, T; \mathbb{R}^d)$ as the standard Euclidean norm in the Euclidean space $\mathbb{R}^m$ or $\mathbb{R}^{m \times d}$ and the set of all $\mathcal{F}_t$ -adapted and mean-square-integrable processes valued in $\mathbb{R}^d$, respectively.

A process $(y_t, z_t) : [0, T] \times \Omega \rightarrow \mathbb{R}^m \times \mathbb{R}^{m \times d}$ is called an L^2 -adapted solution of the BSDE (1.1) if it is $\{\mathcal{F}_t\}$ -adapted, square integrable, and satisfies (1.1) in the sense of

$$(1.2) \quad y_t = \xi + \int_t^T f(s, y_s, z_s) ds - \int_t^T z_s dW_s, \quad t \in [0, T],$$

where $f : [0, T] \times \Omega \times \mathbb{R}^m \times \mathbb{R}^{m \times d} \rightarrow \mathbb{R}^m$ is an adapted stochastic process with respect to $\{\mathcal{F}_t\}$ ($0 \leq t \leq T$) for each (y_t, z_t) , and the third term on the right-hand side is an Itô-type integral. Under certain reasonable regularity conditions for $f(t, y_t, z_t)$, Pardoux and Peng [25] proved existence and uniqueness of the solution of the BSDE (1.2). Some properties and applications of BSDEs are given in [15]. In this paper, we are interested in numerical solution of the BSDE (1.2). Without loss of generality, assume that the BSDE (1.2) admits a unique L^2 -adapted solution (y_t, z_t) .

In this paper, we suppose that the terminal value of y_t is of the form $\varphi(W_T)$. Then the solution (y_t, z_t) of (1.2) (see [15, 20, 27]) can be represented as

$$(1.3) \quad y_t = u(t, W_t), \quad z_t = \nabla u(t, W_t) \quad \forall t \in [0, T],$$

where ∇u denotes the gradient of $u(t, x)$ with respect to the spatial variable x and $u(t, x)$ is the solution of the following parabolic PDE:

$$(1.4) \quad \frac{\partial u}{\partial t} + \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 u}{\partial x_i^2} + f(t, u, \nabla u) = 0$$

with the terminal condition $u(T, x) = \varphi(x)$.

It is well known that when the functions f and φ are bounded and smooth enough with bounded derivatives, the PDE (1.4) has a unique solution $u(t, x)$ which is also bounded and smooth with bounded derivatives (see [16, 30]). Consequently the BSDE (1.2) with $\xi = \varphi(W_T)$ has a unique solution (y_t, z_t) which takes the form (1.3).

Now we introduce some notation which will be used in what follows. Let $\mathcal{F}_s^{t,x}(t \leq s \leq T)$ be a σ -field generated by the Brownian motion $\{x + W_r - W_t, t \leq r \leq s\}$ starting from the time-space point (t, x) , and set $\mathcal{F}^{t,x} = \mathcal{F}_T^{t,x}$. Define $\mathbb{E}[X]$ as the mathematical expectation of the random variable X , and $\mathbb{E}_s^{t,x}[X]$ as the conditional mathematical expectation of the random variable X under the σ -field $\mathcal{F}_s^{t,x}(t \leq s \leq T)$ (i.e., $\mathbb{E}_s^{t,x}[X] = \mathbb{E}[X | \mathcal{F}_s^{t,x}]$). When $s = t$, we simply use $\mathbb{E}_t^x[X]$ to denote $\mathbb{E}[X | \mathcal{F}_t^{t,x}]$. Denote by $\mathbb{Z}$ the set of all integer numbers. Let $x \in \mathbb{R}^d$ mean $x = (x^1, x^2, \dots, x^d)$ with $x^i \in \mathbb{R}$, $i = 1, 2, \dots, d$, and $\mathbf{i} \in \mathbb{Z}^d$ mean $\mathbf{i} = (i_1, i_2, \dots, i_d)$ with $i_k \in \mathbb{Z}$, $k = 1, 2, \dots, d$. We define $(\cdot)^\top$ as the transpose operator for a vector or matrix.

2. Reference equations for the multistep scheme. In this section, we first derive reference equations for the BSDE with $m = d = 1$, and then discuss the case of general BSDEs.

2.1. The case of $m = d = 1$. We assume $d = 1$ and $m = 1$ now. Let N be a positive integer, and consider the uniform partition of the time interval $[0, T]$:

$$(2.1) \quad 0 = t_0 < t_1 < \dots < t_N = T$$

with $t_i = t_0 + i\Delta t$ ($i = 0, 1, \dots, N$) and the time step $\Delta t = \frac{T}{N}$. Assume $(y_t, z_t) : [0, T] \times \Omega \rightarrow \mathbb{R} \times \mathbb{R}$ is the exact solution of the BSDE (1.2). For two given positive integers k and K_y satisfying $1 \leq k \leq K_y \leq N$, it is easy to obtain

$$(2.2) \quad y_{t_n} = y_{t_{n+k}} + \int_{t_n}^{t_{n+k}} f(s, y_s, z_s) ds - \int_{t_n}^{t_{n+k}} z_s dW_s.$$

Taking the conditional mathematical expectation $\mathbb{E}_{t_n}^x[\cdot]$ on both sides of (2.2), we get

$$(2.3) \quad y_{t_n} = \mathbb{E}_{t_n}^x[y_{t_{n+k}}] + \int_{t_n}^{t_{n+k}} \mathbb{E}_{t_n}^x[f(s, y_s, z_s)] ds.$$

Notice that the integrand $\mathbb{E}_{t_n}^x[f(s, y_s, z_s)]$ on the right-hand side of (2.3) is a deterministic function of s . To approximate the integral in (2.3), any numerical method for computing integrals with deterministic integrands may be used. When the values of (y_t, z_t) on the time levels $t_{n+1}, t_{n+2}, \dots, t_{n+K_y}$ are available, it is often natural to construct an approximation of the integrand $\mathbb{E}_{t_n}^x[f(s, y_s, z_s)]$ in (2.3) based on these values. If implicit methods are used to solve (y_{t_n}, z_{t_n}) , the values on the time level $t = t_n$ may also be used for the approximation.

Here we choose the Lagrange interpolating polynomial $p_{K_y}^{t_n, x}(s)$ based on the support points $(t_{n+i}, \mathbb{E}_{t_n}^x[f(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}})])$, $i = 0, 1, \dots, K_y$, to approximate the deterministic integrand $\mathbb{E}_{t_n}^x[f(s, y_s, z_s)]$ in (2.3). Thus we have

$$(2.4) \quad \int_{t_n}^{t_{n+k}} \mathbb{E}_{t_n}^x[f(s, y_s, z_s)] ds = \int_{t_n}^{t_{n+k}} p_{K_y}^{t_n, x}(s) ds + R_y^n,$$

where the residual

$$R_y^n = \int_{t_n}^{t_{n+k}} \{\mathbb{E}_{t_n}^x[f(s, y_s, z_s)] - p_{K_y}^{t_n, x}(s)\} ds.$$

With the uniform time partition (2.1), the integral $\int_{t_n}^{t_{n+k}} p_{K_y}^{t_n, x}(s) ds$ can be explicitly presented as

$$(2.5) \quad \int_{t_n}^{t_{n+k}} p_{K_y}^{t_n, x}(s) ds = k\Delta t \sum_{i=0}^{K_y} b_{K_y, i}^k \mathbb{E}_{t_n}^x[f(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}})],$$

where

$$(2.6) \quad \begin{aligned} b_{K_y, i}^k &= \frac{1}{k\Delta t} \int_{t_n}^{t_{n+k}} \prod_{\substack{j=0 \\ j \neq i}}^{K_y} \left(\frac{s - t_{n+j}}{t_{n+i} - t_{n+j}} \right) ds \\ &= \frac{1}{k} \int_0^k \prod_{\substack{j=0 \\ j \neq i}}^{K_y} \left(\frac{s-j}{i-j} \right) ds \end{aligned}$$

for $i = 0, 1, \dots, K_y$. Combining (2.3), (2.4), and (2.5), we obtain the following reference equation for solving y_{t_n} :

$$(2.7) \quad y_{t_n} = \mathbb{E}_{t_n}^x[y_{t_{n+k}}] + k\Delta t \sum_{i=0}^{K_y} b_{K_y, i}^k \mathbb{E}_{t_n}^x[f(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}})] + R_y^n.$$

Now let us turn to derive a reference equation for solving z_{t_n} . Let $\Delta W_s = W_s - W_{t_n}$ for $s \geq t_n$; then ΔW_s is a standard Brownian motion with mean zero and variance $\sqrt{s - t_n}$. Let l and K_z be two positive integers satisfying $1 \leq l \leq K_z \leq N$. Multiply both sides of (2.2) (with k replaced by l) by $\Delta W_{t_{n+l}}$, and then take the conditional mathematical expectation $\mathbb{E}_{t_n}^x[\cdot]$ to both sides of the derived equation; we finally obtain by the Itô isometry formula

$$(2.8) \quad 0 = \mathbb{E}_{t_n}^x [y_{t_{n+l}} \Delta W_{t_{n+l}}] + \int_{t_n}^{t_{n+l}} \mathbb{E}_{t_n}^x [f(s, y_s, z_s) \Delta W_s] ds - \int_{t_n}^{t_{n+l}} \mathbb{E}_{t_n}^x [z_s] ds.$$

In the same way as we obtained the reference equation (2.7), we have

$$(2.9) \quad \begin{aligned} & \int_{t_n}^{t_{n+l}} \mathbb{E}_{t_n}^x [f(s, y_s, z_s) \Delta W_s] ds \\ &= l\Delta t \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [f(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}}) \Delta W_{t_{n+i}}] + R_{z1}^n \end{aligned}$$

and

$$(2.10) \quad - \int_{t_n}^{t_{n+l}} \mathbb{E}_{t_n}^x [z_s] ds = -l\Delta t \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [z_{t_{n+i}}] + R_{z2}^n,$$

where $b_{K_z, i}^l$ is defined by (2.6) with the k and K_y replaced by l and K_z , respectively, and

$$\begin{aligned} R_{z1}^n &= \int_{t_n}^{t_{n+l}} \left\{ \mathbb{E}_{t_n}^x [f(s, y_s, z_s) \Delta W_s] - \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [f(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}}) \Delta W_{t_{n+i}}] \right\} ds, \\ R_{z2}^n &= - \int_{t_n}^{t_{n+l}} \left\{ \mathbb{E}_{t_n}^x [z_s] - \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [z_{t_{n+i}}] \right\} ds. \end{aligned}$$

Using the relation (1.3) between the solution (y_t, z_t) of (1.2) and the solution u of (1.4), it is easy to verify that

$$(2.11) \quad \begin{aligned} \mathbb{E}_{t_n}^x [y_{t_{n+l}} \Delta W_{t_{n+l}}] &= \frac{1}{\sqrt{2\pi l\Delta t}} \int_{-\infty}^{\infty} u(t_{n+l}, x+v) v \exp\left(-\frac{v^2}{2l\Delta t}\right) dv \\ &= \frac{l\Delta t}{\sqrt{2\pi l\Delta t}} \int_{-\infty}^{\infty} \frac{\partial u}{\partial x}(t_{n+l}, x+v) \exp\left(-\frac{v^2}{2l\Delta t}\right) dv \\ &= l\Delta t \mathbb{E}_{t_n}^x [z_{t_{n+l}}]. \end{aligned}$$

Now, combining (2.8), (2.9), (2.10), and (2.11), we get the reference equation for solving z_{t_n} as

$$(2.12) \quad \begin{aligned} 0 &= \mathbb{E}_{t_n}^x [z_{t_{n+l}}] + \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [f(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}}) \Delta W_{t_{n+i}}] \\ &\quad - \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [z_{t_{n+i}}] + \frac{1}{l\Delta t} R_z^n \end{aligned}$$

with $R_z^n = R_{z1}^n + R_{z2}^n$. Equations (2.7) and (2.12) are the two reference equations for the BSDE (1.2) with $m = d = 1$.

2.2. The general case. Now let us turn to getting reference equations for general cases. Suppose that the BSDE (1.2) is driven by a standard d -dimensional Brownian motion $W_t = (W_t^1, W_t^2, \dots, W_t^d)^\top$ with W_t^i 's, $i = 1, 2, \dots, d$, being independent standard one-dimensional Brownian motions. Let $(y_t, z_t) : [0, T] \times \Omega \rightarrow \mathbb{R}^m \times \mathbb{R}^{m \times d}$ be the solution of the BSDE (1.2), where $y_t = (y_t^1, y_t^2, \dots, y_t^m)^\top$ and $z_t = (z_t^{i,j})_{m \times d}$. Just by using the same method that we used to obtain the reference equation (2.7), we get

$$(2.13) \quad y_{t_n}^{k'} = \mathbb{E}_{t_n}^x [y_{t_{n+k}}^{k'}] + k\Delta t \sum_{i=0}^{K_y} b_{K_y, i}^k \mathbb{E}_{t_n}^x [f^{k'}(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}})] + R_y^{n, k'},$$

$$k' = 1, 2, \dots, m,$$

where $f^{k'}$ is the k' 'th component of the vector function f and the residual

$$R_y^{n, k'} = \int_{t_n}^{t_{n+k}} \left\{ \mathbb{E}_{t_n}^x [f^{k'}(s, y_s, z_s)] - p_{K_y}^{t_n, x, k'}(s) \right\} ds.$$

We also may rewrite (2.13) in the vector form

$$(2.14) \quad y_{t_n} = \mathbb{E}_{t_n}^x [y_{t_{n+k}}] + k\Delta t \sum_{i=0}^{K_y} b_{K_y, i}^k \mathbb{E}_{t_n}^x [f(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}})] + R_y^n,$$

where $R_y^n = (R_y^{n,1}, R_y^{n,2}, \dots, R_y^{n,m})^\top$. Again similar to the way to obtain the reference equation (2.12), we have

$$(2.15) \quad 0 = \mathbb{E}_{t_n}^x [z_{t_{n+l}}^{i', j'}] + \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [f^{i'}(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}}) \Delta W_{t_{n+i}}^{j'}]$$

$$- \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [z_{t_{n+i}}^{i', j'}] + \frac{1}{l\Delta t} R_z^{n, i', j'},$$

$$i' = 1, 2, \dots, m, \quad j' = 1, 2, \dots, d,$$

where the residual $R_z^{n, i', j'} = R_{z1}^{n, i', j'} + R_{z2}^{n, i', j'}$ with

$$R_{z1}^{n, i', j'} = \int_{t_n}^{t_{n+l}} \left\{ \mathbb{E}_{t_n}^x [f^{i'}(s, y_s, z_s) \Delta W_s^{j'}] - \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [f^{i'}(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}}) \Delta W_{t_{n+i}}^{j'}] \right\} ds,$$

$$R_{z2}^{n, i', j'} = - \int_{t_n}^{t_{n+l}} \left\{ \mathbb{E}_{t_n}^x [z_s^{i', j'}] - \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [z_{t_{n+i}}^{i', j'}] \right\} ds.$$

Written in the matrix form, this becomes

$$(2.16) \quad 0 = \mathbb{E}_{t_n}^x [z_{t_{n+l}}] + \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [f(t_{n+i}, y_{t_{n+i}}, z_{t_{n+i}}) \Delta W_{t_{n+i}}^\top]$$

$$- \sum_{i=0}^{K_z} b_{K_z, i}^l \mathbb{E}_{t_n}^x [z_{t_{n+i}}] + \frac{1}{l\Delta t} R_z^n,$$

where $R_z^n = (R_z^{n,i',j'})_{m \times d}$. Equations (2.14) and (2.16) are the two reference equations for a general BSDE (1.2).

Remark 1.

1. If both the generator function f and the terminal function φ are smooth, then the conditional mathematical expectations $\mathbb{E}_{t_n}^x[f(s, y_s, z_s)]$, $\mathbb{E}_{t_n}^x[f(s, y_s, z_s)\Delta W_s]$, and $\mathbb{E}_{t_n}^x[z_s]$ are also smooth functions of s . Furthermore, by the approximation theory, we have the following estimates for the local error terms R_y^n and R_z^n in (2.14) and (2.16):

$$|R_y^n| = O((\Delta t)^{K_y+2}), \quad |R_z^n| = O((\Delta t)^{K_z+2}).$$

In order to balance the local truncation errors, the positive integers K_y and K_z should satisfy $K_z = K_y + 1$ when the generator f in the BSDE (1.2) explicitly depends on both y and z .

2. In the reference equations (2.14) and (2.16), we have used four positive integers k , K_y , l , and K_z . These four integers could be any positive ones satisfying $1 \leq k \leq K_y \leq N$ and $1 \leq l \leq K_z \leq N$, but the requirement of stability of the multistep scheme will impose some restrictions on the selection of them. We will discuss this in section 3.

3. A stable multistep discretization scheme.

3.1. The semidiscrete scheme. Suppose that $y^n = (y^{1,n}, y^{2,n}, \dots, y^{m,n})^\top$, $z^n = (z^{i,j,n})_{m \times d}$, and $K = \max\{K_y, K_z\}$. We regard y^n as an approximation to y_t , and z^n as an approximation to z_t at the time t_n . Note that $\Delta W_{t_n}^\top = 0$ in (2.16). By omitting the two residual terms R_y^n and R_z^n in (2.14) and (2.16), we propose a multistep semidiscrete scheme for solving the BSDE (1.2) as follows: Given random variables y^{N-i} and z^{N-i} , $i = 0, 1, \dots, K-1$, find random variables y^n and z^n , $n = N-K, \dots, 0$, such that

$$(3.1) \quad \begin{aligned} y^n &= \mathbb{E}_{t_n}^x[y^{n+k}] + k\Delta t \sum_{i=0}^{K_y} b_{K_y,i}^k \mathbb{E}_{t_n}^x[f(t_{n+i}, y^{n+i}, z^{n+i})], \\ 0 &= \mathbb{E}_{t_n}^x[z^{n+l}] + \sum_{i=1}^{K_z} b_{K_z,i}^l \mathbb{E}_{t_n}^x[f(t_{n+i}, y^{n+i}, z^{n+i})\Delta W_{t_{n+i}}^\top] - \sum_{i=0}^{K_z} b_{K_z,i}^l \mathbb{E}_{t_n}^x[z^{n+i}]. \end{aligned}$$

We must choose the positive integers K_y , K_z , k , and l in such a way that the scheme (3.1) for solving y^n and z^n is stable. Setting the generator function $f = 0$ and taking the mathematical expectation $\mathbb{E}[\cdot]$ on both sides of the two equations in (3.1), we obtain

$$(3.2) \quad \mathbb{E}[y^n] = \mathbb{E}[y^{n+k}]$$

and

$$(3.3) \quad 0 = \mathbb{E}[z^{n+l}] - \sum_{i=0}^{K_z} b_{K_z,i}^l \mathbb{E}[z^{n+i}].$$

From (3.2) it is clear that the scheme (3.1) for solving y^n is stable for any positive integers k and K_y ($1 \leq k \leq K_y \leq N$). However, it is often better for the coefficients $b_{K_y,i}^k$ ($i = 0, 1, \dots, K_y$) to be positive and $\sum_{i=0}^{K_y} b_{K_y,i}^k = 1$. If we choose $k = K_y$, it

TABLE 3.1
The coefficients $\{b_{K_y,i}^{K_y}\}_{i=0}^{K_y}$ for $K_y = 1, 2, \dots, 7$ and $K_y = 9$.

K_y	$b_{K_y,i}^{K_y}$									
	$i = 0$	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$	$i = 8$	$i = 9$
1	$\frac{1}{2}$	$\frac{1}{2}$								
2	$\frac{1}{6}$	$\frac{4}{6}$	$\frac{1}{6}$							
3	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$						
4	$\frac{7}{90}$	$\frac{16}{45}$	$\frac{2}{15}$	$\frac{16}{45}$	$\frac{7}{90}$					
5	$\frac{19}{288}$	$\frac{25}{96}$	$\frac{25}{144}$	$\frac{25}{144}$	$\frac{25}{96}$	$\frac{19}{288}$				
6	$\frac{41}{840}$	$\frac{9}{35}$	$\frac{9}{280}$	$\frac{34}{105}$	$\frac{9}{280}$	$\frac{9}{35}$	$\frac{41}{840}$			
7	$\frac{751}{17280}$	$\frac{3577}{17280}$	$\frac{49}{640}$	$\frac{2989}{17280}$	$\frac{2989}{17280}$	$\frac{49}{640}$	$\frac{3577}{17280}$	$\frac{751}{17280}$		
9	$\frac{2857}{89600}$	$\frac{15741}{89600}$	$\frac{27}{2240}$	$\frac{1209}{5600}$	$\frac{2889}{44800}$	$\frac{2889}{44800}$	$\frac{1209}{5600}$	$\frac{27}{2240}$	$\frac{15741}{89600}$	$\frac{2857}{89600}$

is easy to verify that the coefficients $b_{K_y,i}^k$, $i = 0, 1, \dots, K_y$, satisfy such a condition for $K_y = 1, 2, \dots, 7$ and $K_y = 9$; see Table 3.1 for a list of the coefficients $\{b_{K_y,i}^{K_y}\}_{i=0}^{K_y}$. Thus we set $k = K_y$ in our scheme (3.1).

Remark 2. The coefficients $\{b_{K_y,i}^{K_y}\}_{i=0}^{K_y}$ for each K_y listed in Table 3.1 are all positive, and their sum equals 1. From the stability point of view, the sum of absolute values of these coefficients for each fixed K_y should be less than or equal to 1. However, when $K_y = 8$ or $K_y > 9$, some of the coefficients $b_{K_y,i}^{K_y}$, $i = 0, 1, \dots, K_y$, are negative, and some are even bigger than 1. Thus, we excluded the cases of $K_y = 8$ and $K_y > 9$ from Table 3.1.

Equation (3.3) is a difference equation of $\mathbb{E}[z^n]$. Since the procedure for solving $\mathbb{E}[z^n]$ is backward ($n = N - K, \dots, 1, 0$), the characteristic polynomial of the backward difference equation (3.3) is

$$(3.4) \quad \rho_{K_z}^l(\lambda) = \lambda^{K_z-l} - \sum_{i=0}^{K_z} b_{K_z,i}^l \lambda^{K_z-i} = \sum_{i=0}^{K_z} \alpha_{K_z,i}^l \lambda^{K_z-i},$$

where

$$\alpha_{K_z,i}^l = \begin{cases} -b_{K_z,i}^l, & i \neq l, \\ 1 - b_{K_z,i}^l, & i = l. \end{cases}$$

In order for the scheme (3.1) to be stable, roots of the polynomial (3.4) must satisfy the following *root conditions*:

1. The roots of $\rho_{K_z}^l(\lambda)$ must lie on or within a unit circle.
2. The roots of $\rho_{K_z}^l(\lambda)$ on the unit circle must be simple.

Thus, the positive integers K_z and l must be chosen in a way such that the characteristic polynomial $\rho_{K_z}^l(\lambda)$ satisfies the above two conditions. In Table 3.2, we list the coefficients $\alpha_{K_z,i}^l$ ($i = 0, \dots, l$) and the roots $\lambda_{K_z,i}^l$ ($i = 1, \dots, l$) of $\rho_{K_z}^l(\lambda)$ for $K_z = 1, 2, 3$ and $l = 1, \dots, K_z$.

Remark 3. It is worth noting according to Table 3.2 that l can be 1 only for $K_z = 1, 2, 3$ in order to satisfy the above root conditions. The roots of $\rho_{K_z}^l(\lambda)$ do not satisfy the above two root conditions when $l > 1$ or $K_z \geq 4$, as we find.

TABLE 3.2

The coefficients $\{\alpha_{K_z,i}^l\}_{i=0}^l$ and the roots $\{\lambda_{K_z,i}^l\}_{i=0}^l$ of $\rho_{K_z}^l(\lambda)$ for $K_z = 1, 2, 3$ and $l = 1, \dots, K_z$.

K_z	l	$\alpha_{K_z,i}^l$				Roots $\lambda_{K_z,i}^l$		
		$i=0$	$i=1$	$i=2$	$i=3$	$j=1$	$j=2$	$j=3$
1	1	$-\frac{1}{2}$	$\frac{1}{2}$			1		
	2	$-\frac{5}{12}$	$\frac{4}{12}$	$\frac{1}{12}$		1	$-\frac{1}{5}$	
2	1	$-\frac{1}{6}$	$-\frac{4}{6}$	$\frac{5}{6}$		1	-5	
	2	$-\frac{9}{24}$	$\frac{5}{24}$	$\frac{5}{24}$	$-\frac{1}{24}$	1	$-\frac{2}{9} + \frac{\sqrt{13}}{9}$	$-\frac{2}{9} - \frac{\sqrt{13}}{9}$
3	1	$-\frac{1}{6}$	$-\frac{4}{6}$	$\frac{5}{6}$	0	1	0	-5
	2	$-\frac{1}{8}$	$-\frac{3}{8}$	$-\frac{3}{8}$	$\frac{7}{8}$	1	$-2 + \sqrt{3}i$	$-2 - \sqrt{3}i$

Let $K = \max(K_y, K_z)$. By the above stability discussions with data listed in Tables 3.1 and 3.2, we then obtain a stable multistep semidiscrete scheme for solving the BSDE (1.2) as follows: Given random variables y^{N-i} and z^{N-i} , $i = 0, 1, \dots, K-1$, find random variables y^n and z^n , $n = N-K, \dots, 0$, such that

$$(3.5) \quad \begin{aligned} y^n &= \mathbb{E}_{t_n}^x [y^{n+K_y}] + K_y \Delta t \sum_{i=0}^{K_y} b_{K_y,i}^{K_y} \mathbb{E}_{t_n}^x [f(t_{n+i}, y^{n+i}, z^{n+i})], \\ 0 &= \mathbb{E}_{t_n}^x [z^{n+1}] + \sum_{i=1}^{K_z} b_{K_z,i}^1 \mathbb{E}_{t_n}^x [f(t_{n+i}, y^{n+i}, z^{n+i}) \Delta W_{t_{n+i}}^\top] - \sum_{i=0}^{K_z} b_{K_z,i}^1 \mathbb{E}_{t_n}^x [z^{n+i}]. \end{aligned}$$

Remark 4.

1. When $K_y = K_z = 1$, the scheme (3.5) is simply a one-step scheme.
2. According to Remark 1, the two positive integers K_y and K_z in the scheme (3.5) should be related by $K_z = K_y + 1$. When f does not depend on z_t , we may set $K_z = K_y$ to solve y_t more accurately.
3. The scheme (3.5) is globally an implicit scheme, but it is always explicit for solving z^n .

3.2. The fully discrete scheme. The scheme (3.5) is a stable semidiscretization scheme in the time direction. To develop a fully discrete scheme for solving (y^n, z^n) numerically, spatial approximations are also needed. To proceed, we partition the d -dimensional Euclidean space $\mathbb{R}^d$ by

$$(3.6) \quad \mathfrak{R}_h^d = \overbrace{\mathfrak{R}_h^1 \times \mathfrak{R}_h^2 \times \dots \times \mathfrak{R}_h^d}^{d \text{ times}},$$

where $\mathfrak{R}_h^k$ is a partition of the one-dimensional real axis defined by

$$\mathfrak{R}_h^k = \left\{ x_i^k \mid x_i^k \in \mathbb{R}, i \in \mathbb{Z}, x_i^k < x_{i+1}^k, \lim_{i \rightarrow +\infty} x_i^k = +\infty, \lim_{i \rightarrow -\infty} x_i^k = -\infty \right\}$$

for $k = 1, 2, \dots, d$. Let $x_{\mathbf{i}} = (x_{i_1}^1, x_{i_2}^2, \dots, x_{i_d}^d)$ for $\mathbf{i} = (i_1, i_2, \dots, i_d) \in \mathbb{Z}^d$.

Based on the semidiscrete scheme (3.5), we then propose the stable multistep time-space fully discrete scheme for solving the BSDE (1.2) as follows: Given the random variables $y_{\mathbf{i}}^{N-l}$, $\mathbf{i} \in \mathbb{Z}^d$ and $l = 0, 1, \dots, K-1$, find the random variables

(y_i^n, z_i^n) , $i \in \mathbb{Z}^d$ and $n = N - K, \dots, 0$, such that

$$\begin{aligned}
 (3.7) \quad y_i^n &= \hat{\mathbb{E}}_{t_n}^{x_i}[\hat{y}^{n+K_y}] + K_y \Delta t \sum_{j=1}^{K_y} b_{K_y, j}^{K_y} \hat{\mathbb{E}}_{t_n}^{x_i}[f(t_{n+j}, \hat{y}^{n+j}, \hat{z}^{n+j})] \\
 &\quad + K_y \Delta t b_{K_y, 0}^{K_y} f(t_n, y_i^n, z_i^n), \\
 0 &= \hat{\mathbb{E}}_{t_n}^{x_i}[\hat{z}^{n+1}] + \sum_{j=1}^{K_z} b_{K_z, j}^1 \hat{\mathbb{E}}_{t_n}^{x_i}[f(t_{n+j}, \hat{y}^{n+j}, \hat{z}^{n+j}) \Delta W_{t_{n+j}}^\top] \\
 &\quad - \sum_{j=1}^{K_z} b_{K_z, j}^1 \hat{\mathbb{E}}_{t_n}^{x_i}[\hat{z}^{n+j}] - b_{K_z, 0}^1 z_i^n.
 \end{aligned}$$

Here $\hat{y}^{n+1}$ and $\hat{z}^{n+1}$ are interpolating values from $\{y_1^{n+1} \mid \mathbf{1} \in \mathbb{Z}^d\}$ and $\{z_1^{n+1} \mid \mathbf{1} \in \mathbb{Z}^d\}$ at the space points $x_i + W_{t_{n+1}} - W_{t_n}$, respectively, and $\hat{y}^{n+j}$ and $\hat{z}^{n+j}$ are defined for $j = 2, \dots, K_y$ or K_z in a similar way; $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$ is an approximation of the conditional mathematical expectation $\mathbb{E}_{t_n}^{x_i}[\cdot]$. Many numerical methods for computing $\mathbb{E}_{t_n}^{x_i}[\cdot]$ may be used as $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$; for example, $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$ can be a Monte Carlo method or a quadrature rule used to approximate the conditional mathematical expectation $\mathbb{E}_{t_n}^{x_i}[\cdot]$.

Remark 5. The multistep scheme (3.7) is a scheme for solving general BSDEs driven by a d -dimensional Brownian motion with d being any positive integer. From the structure of the scheme, we can see that its numerical solution procedures among space grids are independent, and the information needed to solve the solution at a space-time grid can be localized according to the accuracy required. Thus, parallel computing techniques can be adopted in our scheme to solve large scale problems.

In our numerical tests in section 5, we take $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$ as the approximation of $\mathbb{E}_{t_n}^{x_i}[\cdot]$ using the Gauss–Hermite quadrature rules. A main benefit of this approach is due to the high-order accuracy of the Gauss–Hermite quadratures so that one needs to evaluate the integrand function at only a few points—typically no more than eight points for each of the space dimensions, i.e., no more than 8^d points in total. This will significantly reduce the time spent in computing conditional mathematical expectations compared with Monte Carlo methods, especially when the integrand function f is expensive to calculate.

For a one-dimensional function $g(x)$, the Gauss–Hermite quadrature formula for computing the integral $\int_{-\infty}^{\infty} e^{-x^2} g(x) dx$ is given by

$$(3.8) \quad \int_{-\infty}^{\infty} e^{-x^2} g(x) dx \approx \sum_{i=1}^L w_i g(a_i),$$

where w_i , $i = 1, 2, \dots, L$, are weights defined by

$$w_i = \frac{2^{L+1} L! \sqrt{\pi}}{[H'_L(a_i)]^2},$$

and a_i , $i = 1, 2, \dots, L$, are roots of the Hermite polynomial of degree L ,

$$H_L(x) = (-1)^L e^{x^2} \frac{d^L}{dx^L} (e^{-x^2}).$$

The weights $\{w_i\}_{i=1}^L$ and the roots $\{a_i\}_{i=1}^L$ for different L 's can be found in [1, p. 924]. The truncation error of the Gauss–Hermite quadrature formula is

$$R(g, G_L) = \frac{L! \sqrt{\pi}}{2^L (2L)!} g^{(2L)}(\eta),$$

where η is some real number. For a d -dimensional function $g(x)$, $x \in \mathbb{R}^d$, the Gauss–Hermite quadrature formula is

$$(3.9) \quad \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} e^{-x_1^2 - x_2^2 - \cdots - x_d^2} g(x) dx_1 dx_2 \cdots dx_d \approx \sum_{j_1=1, \dots, j_d=1}^{L, \dots, L} w_{j_1} \cdots w_{j_d} g(a_{j_1}, \dots, a_{j_d}).$$

When $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$ is the approximation of $\mathbb{E}_{t_n}^{x_i}[\cdot]$ using the Gauss–Hermite quadrature rule, we have the following formula:

$$(3.10) \quad \hat{\mathbb{E}}_{t_n}^{x_i}[\hat{y}^{n+k}] = \frac{1}{\pi^{\frac{d}{2}}} \sum_{\mathbf{j}=1}^L w_{\mathbf{j}} \hat{y}^{n+k}(x_{\mathbf{i}} + \sqrt{2k\Delta t} a_{\mathbf{j}}),$$

where $\mathbf{j} = (j_1, j_2, \dots, j_d)$, $w_{\mathbf{j}} = \prod_{i=1}^d w_{j_i}$, $a_{\mathbf{j}} = (a_{j_1}, a_{j_2}, \dots, a_{j_d})$, $\sum_{\mathbf{j}=1}^L = \sum_{j_1=1, \dots, j_d=1}^{L, \dots, L}$. We note that $\hat{y}^{n+k}(x_{\mathbf{i}} + \sqrt{2k\Delta t} a_{\mathbf{j}})$ is the interpolating value at the space point $x_{\mathbf{i}} + \sqrt{2k\Delta t} a_{\mathbf{j}}$ based on the values of y^{n+k} at a finite number of the space grid points in $\mathfrak{R}_h^d$ near $x_{\mathbf{i}} + \sqrt{2k\Delta t} a_{\mathbf{j}}$. Approximate Gauss–Hermite quadrature formulas for computing $\hat{\mathbb{E}}_{t_n}^{x_i}[f(t_{n+k}, \hat{y}^{n+k}, \hat{z}^{n+k})]$, $\hat{\mathbb{E}}_{t_n}^{x_i}[\hat{z}^{n+k}]$, and $\hat{\mathbb{E}}_{t_n}^{x_i}[f(t_{n+k}, \hat{y}^{n+k}, \hat{z}^{n+k})\Delta W_{t_{n+k}}]$ are defined similarly.

Remark 6. In order to solve (y_i^n, z_i^n) , $n = N - K, N - K - 1, \dots, 0$, using the multistep scheme (3.7), the starting values (y^{N-l}, z^{N-l}) , $l = 1, \dots, K - 1$, must be previously given or computed. To do that, some other numerical methods or the scheme (3.7) with $K_z = K_y = 1$ may be used with a smaller time step.

Remark 7. It is worth noting that the Gauss–Hermite quadrature formula can be extended to general Markovian diffusions. When the forward component is a general Markovian diffusion, the diffusion process needs to be approximated by using some methods such as the Euler–Maruyama scheme or the Milstein scheme; then the approximate diffusion is a function of Brownian motion. Thus the density of the Brownian W_t can be used to compute the conditional mathematical expectation $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$.

Remark 8. To compute $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$ using the Gauss–Hermite quadrature rule, some interpolation methods are needed when the quadrature points are not grid points. The number of the Gauss–Hermite quadrature points grows exponentially along the increase of the problem dimension d without extra special treatments, and this is a common challenging problem for most numerical methods for solving high-dimension problems. We think sparse grid integration techniques could be combined to efficiently compute $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$ in our multistep scheme when applied to high-dimensional BSDEs.

4. Error estimates of the multistep semidiscrete scheme for some types of BSDEs. In this section, we will carry out error analysis for the multistep semidiscrete scheme (3.5) for solving the BSDE (1.2) with the generator function f being independent of the random variable z_t , i.e.,

$$(4.1) \quad y_t = \varphi(W_T) + \int_t^T f(s, y_s) ds - \int_t^T z_s dW_s.$$

For simplicity of the analysis, we consider only the case of $m = d = 1$, but all results obtained below also hold for general m and d . In such a simplified case, the reference

equations (2.7) and (2.12) with $k = K_y$ and $l = 1$ have the following reduced form:
(4.2)

$$\begin{aligned} y_{t_n} &= \mathbb{E}_{t_n}^x [y_{t_{n+K_y}}] + K_y \Delta t \sum_{i=0}^{K_y} b_{K_y,i}^{K_y} \mathbb{E}_{t_n}^x [f(t_{n+i}, y_{t_{n+i}})] + R_y^n, \\ 0 &= \mathbb{E}_{t_n}^x [z_{t_{n+1}}] + \sum_{i=0}^{K_z} b_{K_z,i}^1 \mathbb{E}_{t_n}^x [f(t_{n+i}, y_{t_{n+i}}) \Delta W_{t_{n+i}}] - \sum_{i=0}^{K_z} b_{K_z,i}^1 \mathbb{E}_{t_n}^x [z_{t_{n+i}}] + \frac{1}{\Delta t} R_z^n. \end{aligned}$$

Then the corresponding multistep scheme (3.5) becomes

$$\begin{aligned} (4.3) \quad y^n &= \mathbb{E}_{t_n}^x [y^{n+K_y}] + K_y \Delta t \sum_{i=0}^{K_y} b_{K_y,i}^{K_y} \mathbb{E}_{t_n}^x [f(t_{n+i}, y^{n+i})], \\ 0 &= \mathbb{E}_{t_n}^x [z^{n+1}] + \sum_{i=0}^{K_z} b_{K_z,i}^1 \mathbb{E}_{t_n}^x [f(t_{n+i}, y^{n+i}) \Delta W_{t_{n+i}}] - \sum_{i=0}^{K_z} b_{K_z,i}^1 \mathbb{E}_{t_n}^x [z^{n+i}] \end{aligned}$$

for $n = N - K, \dots, 0$ with $K = \max(K_y, K_z)$.

To simplify the presentation, we assume that the following two assumptions hold.

ASSUMPTION 1. *The positive integers K_y and K_z are chosen such that the roots of the characteristic polynomial*

$$(4.4) \quad \rho_{K_z}^1(\lambda) = \lambda^{K_z-1} - \sum_{i=0}^{K_z} b_{K_z,i}^1 \lambda^{K_z-i}$$

satisfy the root conditions.

ASSUMPTION 2. *The functions $f(t, y_t)$ and $\varphi(W_T)$ are bounded and smooth enough with bounded derivatives.*

Under Assumption 2, it is well known that the parabolic PDE (1.4) with the generator function $f = f(t, u)$ has a unique solution $u(t, x)$ which is also bounded and smooth with bounded derivatives (see [16, 30]), and consequently the BSDE (4.1) has a unique solution (y_t, z_t) which can be represented as that in (1.3).

We first present two lemmas as follows.

LEMMA 1. *Suppose that the functions $g = g(s, x)$ and $v = v(s, x)$ are bounded and smooth with bounded derivatives. Define $G(s) = \mathbb{E}_t^x [g(s, v(s, x + W_s - W_t))]$ and $\hat{G}(s) = \mathbb{E}_t^x [g(s, v(s, x + W_s - W_t))(W_s - W_t)]$, $t \leq s$. Then $G(s)$ and $\hat{G}(s)$ have bounded derivatives.*

LEMMA 2. *Let R_y^n and R_z^n be the local truncation errors defined in the reference equation (4.2). Then under Assumption 2, we have the following local estimates:*

$$(4.5) \quad |R_y^n| \leq C(\Delta t)^{K_y+2}, \quad |R_z^n| \leq C(\Delta t)^{K_z+2},$$

where $C > 0$ is a generic constant depending only on T , the upper bounds of φ and f , and their derivatives.

Proofs of the above two lemmas are similar to those of Lemma 3.1 and 3.2 in [34], and thus we omit them here. A lemma regarding the backward discrete Gronwall inequality is given below.

LEMMA 3. *Suppose that N and K are two nonnegative integers with $N \geq K$ and Δt any positive number. Let $\{\eta_n\}$ be a series satisfying*

$$(4.6) \quad |\eta_n| \leq \beta + \alpha \Delta t \sum_{j=n+1}^N |\eta_j|, \quad n = N - K, N - K - 1, \dots, 0,$$

where α and β are two positive constants. Let $M_0 = \max_{N-K < j \leq N} |\eta_j|$ and $T = N\Delta t$; then

$$(4.7) \quad |\eta_n| \leq e^{\alpha T} (\beta + \alpha K \Delta t M_0), \quad n = N - K, N - K - 1, \dots, 0.$$

Proof. Let $\zeta_n = \beta + \alpha \Delta t \sum_{j=n}^N |\eta_j|$, $n = N - K + 1, N - K, \dots, 0$; then we easily get $|\eta_n| \leq \zeta_{n+1}$. Since

$$\zeta_n = \beta + \alpha \Delta t \sum_{j=n+1}^N |\eta_j| + \alpha \Delta t |\eta_n| \leq \zeta_{n+1} + \alpha \Delta t \zeta_{n+1},$$

we get

$$\begin{aligned} |\eta_n| &\leq \zeta_n \leq (1 + \alpha \Delta t) \zeta_{n+1} \leq \dots \leq (1 + \alpha \Delta t)^{N-n-K+1} \zeta_{N-K+1} \\ &\leq e^{\alpha T} \zeta_{N-K+1} \\ &= e^{\alpha T} \left(\beta + \alpha \Delta t \sum_{j=N-K+1}^N |\eta_j| \right) \\ &\leq e^{\alpha T} (\beta + \alpha K \Delta t M_0), \end{aligned}$$

which completes the proof. $\square$

Now we give an error estimate of $y_{t_n} - y^n$ in the following theorem.

THEOREM 1. *Let y_t and y^n be the solutions of the BSDE (4.1) and the multistep scheme (4.3), respectively. Suppose that Assumption 2 holds and the initial values satisfy $\max_{N-K_y < j \leq N} \mathbb{E}[|y_{t_j} - y^j|] = O((\Delta t)^{K_y+1})$. Then for sufficiently small time step Δt we have*

$$(4.8) \quad \sup_{0 \leq n \leq N} \mathbb{E}[|y_{t_n} - y^n|] \leq C(\Delta t)^{K_y+1},$$

where $C > 0$ is a generic constant depending only on T , the upper bounds of functions φ and f , and their derivatives.

Proof. Let $e_y^n = y_{t_n} - y^n$ for $n = N, N - 1, \dots, 0$. From (4.2) and (4.3) we get

$$(4.9) \quad e_y^n = \mathbb{E}_{t_n}^x [e_y^{n+K_y}] + K_y \Delta t \sum_{i=0}^{K_y} b_{K_y, i}^{K_y} \mathbb{E}_{t_n}^x [f(t_{n+i}, y_{t_{n+i}}) - f(t_{n+i}, y^{n+i})] + R_y^n.$$

Then we have

$$(4.10) \quad |e_y^n| \leq \mathbb{E}_{t_n}^x [|e_y^{n+K_y}|] + L K_y \Delta t \sum_{i=0}^{K_y} b_{K_y, i}^{K_y} \mathbb{E}_{t_n}^x [|e_y^{n+i}|] + |R_y^n|,$$

where L is the Lipschitz constant of the function $f(t, y_t)$ with respect to y_t . Taking the mathematical expectation $\mathbb{E}[\cdot]$ on both sides of the inequality (4.10), we obtain

$$(4.11) \quad \mathbb{E}[|e_y^n|] \leq \mathbb{E}[|e_y^{n+K_y}|] + L K_y \Delta t \sum_{i=0}^{K_y} b_{K_y, i}^{K_y} \mathbb{E}[|e_y^{n+i}|] + \mathbb{E}[|R_y^n|].$$

Let $N_K = \lfloor \frac{N-n}{K_y} \rfloor$. For an integer s satisfying $1 \leq s \leq N_K$, we similarly have the estimate

$$(4.12) \quad \mathbb{E}[|e_y^{n+(s-1)K_y}|] \leq \mathbb{E}[|e_y^{n+sK_y}|] + L K_y \Delta t \sum_{i=0}^{K_y} b_{K_y, i}^{K_y} \mathbb{E}[|e_y^{n+(s-1)K_y+i}|] + \mathbb{E}[|R_y^{n+(s-1)K_y}|].$$

Now we add up the above inequalities (4.12) over $s = 1, 2, \dots, N_K$ and obtain

$$(4.13) \quad \mathbb{E}[|e_y^n|] \leq \mathbb{E}[|e_y^{n+N_K K_y}|] + 2LK_y \Delta t \sum_{i=0}^{N_K K_y} \mathbb{E}[|e_y^{n+i}|] + \sum_{i=0}^{N_K-1} \mathbb{E}[|R_y^{n+iK_y}|],$$

which is equivalent to

$$(4.14) \quad (1 - 2LK_y \Delta t) \mathbb{E}[|e_y^n|] \leq \mathbb{E}[|e_y^{n+N_K K_y}|] + 2LK_y \Delta t \sum_{i=n+1}^{N_K K_y} \mathbb{E}[|e_y^i|] + \sum_{i=0}^{N_K-1} \mathbb{E}[|R_y^{n+iK_y}|].$$

Suppose that $D = \frac{1}{1-2LK_y \Delta t}$, $N_1 = \frac{2LK_y}{1-2LK_y \Delta t}$, and $M_0 = \max_{N-K_y < i \leq N} \mathbb{E}[|e_y^i|] + \sum_{i=0}^{N_K-1} \mathbb{E}[|R_y^{n+iK_y}|]$. For sufficiently small time step Δt , D and N_1 are clearly positive and bounded by a positive constant number. Then by Lemma 3 and (4.14), we obtain the following inequality:

$$(4.15) \quad \mathbb{E}[|e_y^n|] \leq e^{N_1 T} (DM_0 + \Delta t K_y N_1 M_0) = e^{N_1 T} (D + \Delta t K_y N_1) M_0.$$

On the other hand, Lemma 2 tells us that

$$\mathbb{E}[|R_y^{n+iK_y}|] \leq C(\Delta t)^{K_y+2},$$

which implies

$$\sum_{i=0}^{N_K-1} \mathbb{E}[|R_y^{n+iK_y}|] \leq N \mathbb{E}[|R_y^{n+iK_y}|] \leq CN(\Delta t)^{K_y+2} = C(\Delta t)^{K_y+1}.$$

The last equality comes from $N\Delta t = T$. By the given condition $\max_{N-K_y < i \leq N} \mathbb{E}[|e_y^i|] = O((\Delta t)^{K_y+1})$ in the theorem, we obtain

$$(4.16) \quad M_0 \leq C(\Delta t)^{K_y+1}.$$

Combining (4.15) and (4.16), we immediately get (4.8), and the proof is now completed. $\square$

Now we turn to estimate $z_{t_n} - z^n$, where z_{t_n} is the solution z_t of the BSDE (4.1) at the time level $t = t_n$, and where z^n is the solution of the multistep scheme (4.3). In order to estimate $z_{t_n} - z^n$, we need to measure the bound of ∇y_t , which satisfies the following variational equation:

$$(4.17) \quad \nabla y_t = \varphi'_x(W_T) + \int_t^T f'_y(s, y_s) \nabla y_s ds - \int_t^T \nabla z_s dW_s,$$

where the Brownian motion W_s starts from x at time t , and the processes ∇y_s and ∇z_s are the variations of y_s and z_s , respectively, with respect to the spatial variable x . For the boundedness of y_t and ∇y_t , we have the following statement under Assumption 2.

LEMMA 4. *Suppose that Assumption 2 holds, and let y_t and ∇y_t be the solutions of (4.1) and (4.17), respectively. Then it holds that*

$$(4.18) \quad \sup_{t \in [0, T]} \mathbb{E}[|y_t|] + \sup_{t \in [0, T]} \mathbb{E}[|\nabla y_t|] \leq C,$$

where $C > 0$ is a generic constant depending only on T , the upper bounds of functions φ and f , and their derivatives.

It is easy to verify that the variation process ∇y^n of the solution y^n of the multistep scheme (4.3) satisfies

$$(4.19) \quad \nabla y^n = \mathbb{E}_{t_n}^x [\nabla y^{n+1}] + \Delta t \sum_{i=0}^{K_y} b_{K_y, i}^1 \mathbb{E}_{t_n}^x [f'_y(t_{n+i}, y^{n+i}) \nabla y^{n+i}]$$

for $n = N - K_y, \dots, 0$ with $\nabla y^N = \varphi'_x(W_T)$. We note here that the Brownian motion W_s starts from x at the time $t = t_n$, and ∇y^n is the variation of y^n with respect to x .

Based on (4.17), we get

$$(4.20) \quad \begin{aligned} \nabla y_{t_n} &= \mathbb{E}_{t_n}^x [\nabla y_{t_{n+1}}] + \int_{t_n}^{t_{n+1}} \mathbb{E}_{t_n}^x [f'_y(s, y_s) \nabla y_s] ds \\ &= \mathbb{E}_{t_n}^x [\nabla y_{t_{n+1}}] + \Delta t \sum_{i=0}^{K_y} b_{K_y, i}^1 \mathbb{E}_{t_n}^x [f'_y(t_{n+i}, y_{t_{n+i}}) \nabla y_{t_{n+i}}] + R_{\nabla y}^n, \end{aligned}$$

where

$$R_{\nabla y}^n = \int_{t_n}^{t_{n+1}} \left\{ \mathbb{E}_{t_n}^x [f'_y(s, y_s) \nabla y_s] - \Delta t \sum_{i=0}^{K_y} b_{K_y, i}^1 \mathbb{E}_{t_n}^x [f'_y(t_{n+i}, y_{t_{n+i}})] \right\}.$$

In the same way as we obtained the estimate of $|R_y^n|$ in Lemma 2, under Assumption 2 we have the following local estimate:

$$(4.21) \quad |R_{\nabla y}^n| \leq Ch^{K_y+2}$$

for $n = N - K_y, \dots, 1, 0$.

Applying the same proof procedure as that used in Theorem 1, we have the following lemma.

LEMMA 5. Suppose that Assumption 2 holds, and let ∇y_t and ∇y^n be the solutions of (4.17) and (4.19), respectively. Then for sufficient small time step Δt , under the condition of the initial approximation $\max_{N-K_y < j \leq N} \mathbb{E}[|\nabla y_{t_j} - \nabla y^j|] = O((\Delta t)^{K_y+1})$, we have

$$(4.22) \quad \sup_{0 \leq n \leq N} \mathbb{E}[|\nabla y_{t_n} - \nabla y^n|] \leq C(\Delta t)^{K_y+1},$$

where $C > 0$ is a generic constant depending only on T , the upper bounds of functions φ and f , and their derivatives.

In the following theorem we present an estimation of $z_{t_n} - z^n$.

THEOREM 2. Let z_t and z^n be the solutions of (4.1) and (4.3), respectively. Suppose that Assumptions 1 and 2 and the conditions of Theorem 1 and Lemma 5 hold, and that the initial values satisfy $\max_{N-K_z < j \leq N} |\mathbb{E}[z_{t_j} - z^j]| = O((\Delta t)^{K_z})$. Then for sufficiently small time step Δt , it holds that

$$(4.23) \quad \sup_{0 \leq n \leq N} \mathbb{E}[|z_{t_n} - z^n|] \leq C(\Delta t)^{\min(K_y+1, K_z)},$$

where $C > 0$ is a generic constant depending only on T , the upper bounds functions φ and f , and their derivatives.

Proof. Let $e_z^n = z_{t_n} - z^n$ for $n = N, N-1, \dots, 1, 0$. From the definition of the conditional mathematical expectation $\mathbb{E}_{t_n}^x[\cdot]$, it is easy to deduce

$$(4.24) \quad \begin{aligned} \mathbb{E}_{t_n}^x[f(t_{n+i}, y_{t_{n+i}})\Delta W_{t_{n+i}}] &= i\Delta t \mathbb{E}_{t_n}^x[f'_y(t_{n+i}, y_{t_{n+i}})\nabla y_{t_{n+i}}], \\ \mathbb{E}_{t_n}^x[f(t_{n+i}, y^{n+i})\Delta W_{t_{n+i}}] &= i\Delta t \mathbb{E}_{t_n}^x[f'_y(t_{n+i}, y^{n+i})\nabla y^{n+i}]. \end{aligned}$$

Combining (4.2), (4.3), and (4.24), we get

$$(4.25) \quad \begin{aligned} 0 &= \mathbb{E}_{t_n}^x[e_z^{n+1}] + \sum_{i=0}^{K_z} b_{K_z, i}^1 \Delta t \mathbb{E}_{t_n}^x[f'_y(t_{n+i}, y_{t_{n+i}})\nabla y_{t_{n+i}} - f'_y(t_{n+i}, y^{n+i})\nabla y^{n+i}] \\ &\quad - \sum_{i=0}^{K_z} b_{K_z, i}^1 \mathbb{E}_{t_n}^x[e_z^{n+i}] + \frac{1}{\Delta t} R_z^n. \end{aligned}$$

Taking mathematical expectation $\mathbb{E}[\cdot]$ on both sides of (4.25), we obtain

$$(4.26) \quad \sum_{i=0}^{K_z} b_{K_z, i}^1 \mathbb{E}[e_z^{n+i}] - \mathbb{E}[e_z^{n+1}] = B_n,$$

where

$$B_n = \sum_{i=0}^{K_z} b_{K_z, i}^1 i \Delta t \mathbb{E}[f'_y(t_{n+i}, y_{t_{n+i}})\nabla y_{t_{n+i}} - f'_y(t_{n+i}, y^{n+i})\nabla y^{n+i}] + \frac{1}{\Delta t} \mathbb{E}[R_z^n].$$

Under Assumption 2, by using Theorem 1 and Lemmas 4 and 5, we get

$$(4.27) \quad \mathbb{E}[|f'_y(t_{n+i}, y_{t_{n+i}}) - f'_y(t_{n+i}, y^{n+i})|] \leq L \mathbb{E}[|e_y^{n+i}|] \leq C(\Delta t)^{K_y+1},$$

and then

$$(4.28) \quad \begin{aligned} &\mathbb{E}[|f'_y(t_{n+i}, y_{t_{n+i}})\nabla y_{t_{n+i}} - f'_y(t_{n+i}, y^{n+i})\nabla y^{n+i}|] \\ &\leq \mathbb{E}[|f'_y(t_{n+i}, y_{t_{n+i}}) - f'_y(t_{n+i}, y^{n+i})| |\nabla y_{t_{n+i}}|] \\ &\quad + \mathbb{E}[|f'_y(t_{n+i}, y^{n+i})| |\nabla y_{t_n}^{n+1} - \nabla y^{n+i}|] \\ &\leq C(\Delta t)^{K_y+1}. \end{aligned}$$

Then the above inequality and Lemma 2 give us an estimation of B_n as

$$(4.29) \quad |B_n| \leq C[(\Delta t)^{K_y+2} + (\Delta t)^{K_z+1}].$$

Equation (4.26) is a difference equation. Let $M'_0 = \max_{N-K_z < i \leq N} \mathbb{E}[|e_z^i|]$. According to Assumption 1 that the roots of its characteristic polynomial (4.4) satisfy root conditions, we obtain (from the theory of finite difference equations for solving ODEs [5, 13, 14])

$$(4.30) \quad \begin{aligned} \mathbb{E}[|e_z^n|] &\leq \bar{D} M'_0 + \bar{M} \sum_{k=0}^{N-K_z} |B_{n+k}| \\ &\leq \bar{D} M'_0 + \bar{M} N \max_{1 \leq j \leq N} |B_j| \\ &\leq \bar{D} (\Delta t)^{K_z} + \bar{M} N [(\Delta t)^{K_y+2} + (\Delta t)^{K_z+1}] \\ &\leq C(\Delta t)^{\min(K_y+1, K_z)}, \end{aligned}$$

where $\bar{D}$ and $\bar{M}$ are some generic positive constants. The proof is completed. $\square$

5. Numerical experiments. In this section, we perform various numerical tests to demonstrate high effectiveness and accuracy of the proposed stable multistep fully discrete scheme (3.7).

Let us first introduce some notation used in what follows. Denote by W_t the standard d -dimensional Brownian motion. We take uniform partitions in both time and space in the experiments, with the time and space step sizes being Δt and Δx , respectively. The time partition number N is then given by $N = \frac{T}{\Delta t}$, where T is the finite terminal time. In the following tables, the notations of $|\cdot|$ and CR represent the standard Euclidean norm in the Euclidean space $\mathbb{R}^m$ or $\mathbb{R}^{m \times d}$ and the convergence rate with respect to the time step Δt , respectively. The convergence rate CR is obtained by using linear least squares fitting. We denote by EMC and θ -MC, respectively, the implicit Euler Monte Carlo method, and the θ -scheme (specifically $\theta = 1/2$) Monte Carlo method introduced in [33], and denote by EGH and θ -GH, respectively, the implicit Euler scheme and the θ -scheme discussed in [34], where the Monte Carlo method is replaced by the Gauss–Hermite quadrature rule for computing conditional mathematical expectations. In the following tests, cubic polynomial interpolations are used for evaluating values at nongrid points during computing conditional mathematical expectations.

The errors $e_y^n = y_{t_n}(x_i) - y_i^n$ and $e_z^n = z_{t_n}(x_i) - z_i^n$ often arise from three sources: the time discretization for obtaining the semidiscrete scheme (3.1), the spatial interpolation for computing $\hat{y}^{n+j}$ and $\hat{z}^{n+j}$, and approximations of the conditional mathematical expectations $\mathbb{E}_{t_n}^{x_i}[\cdot]$ by $\hat{\mathbb{E}}_{t_n}^{x_i}[\cdot]$ using the Gauss–Hermite quadrature formula. In order to obtain optimal numerical solutions, the errors from the three sources often should be balanced. Our main goal is to test the accuracy and convergence rate of the proposed scheme (3.7) with respect to the time step size. To do these, we set the number of the Gauss–Hermite quadrature points to be quite big, such as $L = 8$ in (3.9), so that the error contributed by the use of the Gauss–Hermite quadrature rule is very small and affects measurements of the convergence rate CR very little.

It is well known that the global error of the k -step schemes for solving ODEs is $(\Delta t)^k$ in general. The reference equations (2.7) and (2.12) tell us that the local truncation errors for solving y_t and z_t are $O((\Delta t)^{K_y+2})$ and $O((\Delta t)^{K_z+1})$, respectively. Hence it is expected that the error of the semidiscrete scheme (3.5) is $O((\Delta t)^{K_y+1} + (\Delta t)^{K_z})$ for solving a general nonlinear BSDE with its generator $f = f(t, y_t, z_t)$.

In the fully discrete scheme, the space interpolation error is often $\hat{y}^{n+j} - y^{n+j} = C_x(\Delta x)^r$ with some number $r > 0$, which depends on the interpolation method used and regularity of the functions f and φ with respect to space variables. For the cubic polynomial interpolation, it is clear that $r = 4$. In order to balance errors in the time and space directions, we adjust the space step size Δx and the time step size Δt in the following numerical tests such that they satisfy the equality $(\Delta x)^r = (\Delta t)^{q+1}$, where $q = \min(K_y + 1, K_z)$ with K_y and K_z .

Example 1. The first BSDE we test is

$$(5.1) \quad \begin{cases} -dy_t = (-y_t^3 + 2.5y_t^2 - 1.5y_t)dt - z_t dW_t, \\ y_T = \frac{\exp(W_T + T)}{\exp(W_T + T) + 1}. \end{cases}$$

The analytic solution of (5.1) is given by

$$(5.2) \quad \begin{cases} y_t = \frac{\exp(W_t + t)}{\exp(W_t + t) + 1}, \\ z_t = \frac{\exp(W_t + t)}{(\exp(W_t + t) + 1)^2}. \end{cases}$$

TABLE 5.1
Errors and convergence rates for Example 1.

$ y_0 - y_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EGH	1.517E-02	7.418E-03	3.660E-03	1.817E-03	9.055E-04	1.016
θ -GH	2.241E-04	7.011E-05	1.850E-05	3.793E-06	1.081E-06	1.960
$K_y = 1, K_z = 1, 2$	2.241E-04	7.011E-05	1.850E-05	3.793E-06	1.081E-06	1.960
$K_y = 2, K_z = 1, 2, 3$	4.956E-05	5.550E-06	4.151E-07	5.788E-08	4.894E-09	3.320
$K_y = 3, K_z = 1, 2, 3$	1.360E-06	5.838E-08	2.905E-09	6.406E-10	5.882E-12	4.547
$ z_0 - z_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EGH	1.835E-02	9.138E-03	4.581E-03	2.298E-03	1.149E-03	0.999
θ -GH	2.880E-04	1.658E-04	4.446E-05	7.545E-06	2.645E-06	1.800
$K_y = 1, k_z = 1$	7.329E-04	3.884E-04	2.115E-04	1.127E-04	5.744E-05	0.913
$K_y = 1, k_z = 2$	1.119E-03	3.380E-04	9.117E-05	2.374E-05	6.064E-06	1.889
$K_y = 2, K_z = 1$	1.081E-03	4.994E-04	2.423E-04	1.194E-04	5.933E-05	1.044
$K_y = 2, K_z = 2$	1.157E-03	3.334E-04	9.216E-05	2.420E-05	6.208E-06	1.887
$K_y = 2, K_z = 3$	3.882E-04	7.683E-05	1.179E-05	1.637E-06	2.158E-07	2.758
$K_y = 3, K_z = 1$	1.213E-03	5.855E-04	2.686E-04	1.266E-04	6.119E-05	1.083
$K_y = 3, K_z = 2$	9.721E-04	3.270E-04	9.188E-05	2.423E-05	6.215E-06	1.833
$K_y = 3, K_z = 3$	3.871E-04	7.659E-05	1.179E-05	1.636E-06	2.157E-07	2.817

Note that the generator f in this example does not contain the component z_t . We set the terminal time $T = 1$ and numerically solve (5.1) by using EGH, θ -GH, and the stable multistep fully discrete scheme (3.7). In particular, the scheme (3.7) is tested with different step numbers K_z and K_y . The time partition numbers are set to be $N = 4, 8, \dots, 64$.

The exact solution of (5.1) at $t = 0$, (y_0, z_0) , is $(\frac{1}{2}, \frac{1}{4})$ since $W_0 = 0$. Results on the errors between (y_0, z_0) and the numerical solution (y_0^0, z_0^0) , $|y_0 - y_0^0|$, and $|y_0 - z_0^0|$ are given in Table 5.1. Corresponding convergence rates are also presented there.

Remark 9. The aim of Example 1 is to numerically investigate the accuracy of the scheme (3.7) for solving a BSDE with its generator f being independent of the variable z_t , that is, $f = f(t, y_t)$. In this case, it is obvious that computing the approximate solution y_i^n does not depend on the solution of z_i^n in the scheme (3.7) (i.e, if we are interested only in solving y_t , then we do not need to solve z_t), so the accuracy of solving y_t is not affected by the accuracy of solving z_t . We also note for this example that the θ -GH method is equivalent to our multistep scheme (3.7) with $K_y = 1$ in solving y_t , but they are different in solving z_t .

Table 5.1 shows that our multistep scheme (3.7) is a highly effective and accurate one. For instance, both solution errors of the scheme (3.7) on y_t and z_t are already much under 10^{-3} even with $N = 4$ with $K_y = K_z = 1$, while the EGH method needs $N = 64$ for its error on y_t to reach just under 10^{-3} and that on z_t still does not. The solution error for solving y_t is around 10^{-6} for the θ -GH method with $N = 64$, but the time partition number just needs to be $N = 16$ for our scheme (3.7) with $K_y = 2$

or $N = 8$ with $K_y = 3$ to obtain similar or better accuracy; the solution error on z_t is 2.645E-6 for the θ -GH method with $N = 64$, and our scheme (3.7) needs $N = 32$ with $K_y = 2$ and $K_z = 3$ to obtain a smaller error on z_t , but our scheme obtains much better accuracy when solving y_t at the same time (errors of 5.788E-8 versus 1.081E-6).

For this example, the convergence rate of the scheme (3.7) for solving y_t depends only on the step numbers K_y , but that for solving z_t depends on both K_y and K_z according to our analysis in section 4. As shown in Table 5.1, convergence rates for solving y_t and z_t are roughly $K_y + 1$ and K_z , respectively, in this example (slightly larger than $K_y + 1$ in solving y_t for $K_y = 2, 3$ and a little smaller than K_z for $K_z = 2, 3$, partially due to errors in the spatial direction caused by Gaussian-Hermite quadratures and cubic interpolations). These observations are consistent with the theoretical results presented in Theorems 1 and 2. Thus, in order to obtain the same accuracy in solving y_t and z_t and best efficiency when the generator f of a BSDE is independent of z_t , we should set $K_z = K_y + 1$ in the scheme (3.7).

Example 2. Consider the nonlinear BSDE

$$(5.3) \quad \begin{cases} -dy_t = \frac{1}{2}[\exp(t^2) - 4ty_t - 3\exp(t^2 - y_t \exp(-t^2)) + z_t^2 \exp(-t^2)]dt - z_t dW_t, \\ y_T = \ln(\sin W_T + 3) \exp(T^2), \end{cases}$$

where the generator f is nonlinear and depends on t , y_t , and z_t explicitly. The analytic solution of (5.3) is

$$(5.4) \quad \begin{cases} y_t = \ln(\sin W_t + 3) \exp(t^2), \\ z_t = \exp(t^2) \frac{\cos W_t}{\sin W_t + 3}. \end{cases}$$

Let the terminal time $T = 1$ again. Then the exact solution (y_t, z_t) at the time $t = 0$ is $(y_0, z_0) = (\ln 3, \frac{1}{3})$. We solve (y_0^0, z_0^0) by using EGH, θ -GH, and the scheme (3.7) with different step numbers K_y and K_z , respectively. The time partition number N is again taken to be 4, 8, ..., 64. Convergence rates and errors between (y_0, z_0) and the numerical solution (y_0^0, z_0^0) are given in Table 5.2.

Remark 10. Numerical results for Example 2 listed in Table 5.2 show that the multistep scheme (3.7) still works very well for solving general nonlinear BSDEs and is much more effective than the EGH and θ -GH methods. A significant difference between Examples 2 and 1 is that the accuracy of the multistep scheme in solving z_t affects that in solving y_t in Example 2. The convergence rates of the scheme (3.7) for solving the y_t and z_t are both roughly $\min(K_y + 1, K_z)$ (slightly smaller), as shown in Table 5.2, which is consistent with the local truncation errors discussed in section 2. Thus, in order to obtain the best efficiency when solving general BSDEs, it is better to set $K_z = K_y + 1$ in the scheme (3.7).

Example 3. Let $W_t = \begin{pmatrix} W_t^1 \\ W_t^2 \end{pmatrix}$ be a two-dimensional Brownian motion, where W_t^1 and W_t^2 are two independent standard one-dimensional Brownian motions. We consider a BSDE driven by such W_t as follows:

$$(5.5) \quad \begin{cases} -dy_t = \left[\frac{3}{2}y_t + \ln((z_t \vec{A})^2 + y_t^2) - 2tz_t \vec{A} \right] dt - z_t dW_t, \\ y_T = \sin(\vec{M}W_T + T^2) \exp(-(\vec{N}W_T)^2), \end{cases}$$

TABLE 5.2
Errors and convergence rates for Example 2.

$ y_0 - y_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EGH	6.367E-01	3.256E-01	1.652E-01	8.333E-02	4.185E-02	0.982
θ -GH	6.898E-02	1.686E-02	4.185E-03	1.046E-03	2.621E-04	2.009
$K_y = 1, K_z = 1$	5.345E-02	2.538E-02	1.239E-02	6.136E-03	3.056E-03	1.031
$K_y = 1, K_z = 2$	2.833E-02	1.053E-02	3.171E-03	8.682E-04	2.270E-04	1.753
$K_y = 2, K_z = 1$	6.754E-03	2.837E-03	1.717E-03	9.720E-04	5.180E-04	0.895
$K_y = 2, K_z = 2$	2.211E-03	4.199E-04	1.899E-04	5.368E-05	1.466E-05	1.744
$K_y = 2, K_z = 3$	8.717E-04	2.251E-04	3.580E-05	4.855E-06	6.237E-07	2.643
$K_y = 3, K_z = 1$	3.413E-03	1.657E-03	1.331E-03	8.541E-04	4.853E-04	0.758
$K_y = 3, K_z = 2$	2.437E-03	2.151E-04	1.193E-04	4.476E-05	1.289E-05	1.739
$K_y = 3, K_z = 3$	2.997E-03	2.606E-04	4.895E-05	5.224E-06	6.407E-07	3.002

$ z_0 - z_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EGH	1.548E-02	6.134E-03	2.642E-03	1.238E-03	6.028E-04	1.167
θ -GH	2.156E-02	5.583E-03	1.402E-03	3.514E-04	8.837E-05	1.985
$K_y = 1, K_z = 1$	5.345E-02	2.538E-02	1.239E-02	6.136E-03	3.056E-03	1.031
$K_y = 1, K_z = 2$	1.306E-02	3.933E-03	1.095E-03	2.894E-04	7.444E-05	1.868
$K_y = 2, K_z = 1$	3.028E-02	1.921E-02	1.075E-02	5.680E-03	2.918E-03	0.851
$K_y = 2, K_z = 2$	1.199E-02	3.476E-03	9.417E-04	2.442E-04	6.250E-05	1.900
$K_y = 2, K_z = 3$	4.842E-03	1.066E-03	1.642E-04	2.268E-05	2.981E-06	2.689
$K_y = 3, K_z = 1$	1.634E-02	1.428E-02	9.258E-03	5.267E-03	2.809E-03	0.752
$K_y = 3, K_z = 2$	4.848E-03	2.237E-03	7.567E-04	2.201E-04	5.935E-05	1.605
$K_y = 3, K_z = 3$	4.491E-03	1.045E-03	1.622E-04	2.247E-05	2.960E-06	2.767

where $z_t = (z_t^1, z_t^2)$, $\vec{A} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\vec{M} = (1, 0)$, and $\vec{N} = (0, 1)$. Note that the generator f here is nonlinear and again depends on t , y_t , and z_t explicitly.

The analytic solution of (5.5) is given by

$$(5.6) \quad \begin{cases} y_t = \sin(\vec{M}W_t + t^2) \exp(-(\vec{N}W_t)^2), \\ z_t = \begin{pmatrix} \cos(\vec{M}W_t + t^2) \exp(-(\vec{N}W_t)^2) \\ -2(\vec{N}W_t) \sin(\vec{M}W_t + t^2) \exp(-(\vec{N}W_t)^2) \end{pmatrix}^\top. \end{cases}$$

Let the terminal time $T = 1$. Then the exact solution (y_t, z_t) at the time $t = 0$ is $(y_0, z_0) = (0, (1, 0))$. We solve (y_0^0, z_0^0) by using EGH, θ -GH, and the scheme (3.7) with different step numbers K_y and K_z , respectively. The time partition number N is again taken to be $4, 8, \dots, 64$. Convergence rates and errors between (y_0, z_0) and the numerical solution (y_0^0, z_0^0) are given in Table 5.3.

Remark 11. Example 3 tests the ability of our multistep scheme (3.7) to solve a two-dimensional BSDE (i.e., for $d = 2$, the equation is driven by two independent standard one-dimensional Brownian motions). Numerical results in Table 5.3 show

TABLE 5.3
Errors and convergence rates for Example 3.

$ y_0 - y_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EGH	6.692E-01	3.415E-01	1.731E-01	8.727E-02	4.382E-02	0.983
θ -GH	7.053E-02	1.741E-02	2.897E-03	7.263E-04	1.817E-04	2.006
$K_y = 1, K_z = 1$	8.617E-02	2.460E-02	1.036E-02	4.604E-03	1.968E-03	1.332
$K_y = 1, K_z = 2$	2.216E-02	3.338E-03	1.097E-03	4.593E-04	1.464E-04	1.735
$K_y = 2, K_z = 1$	6.316E-02	7.802E-03	3.978E-03	2.080E-03	1.536E-03	1.232
$K_y = 2, K_z = 2$	3.051E-02	7.935E-03	2.237E-03	3.014E-04	9.346E-05	2.142
$K_y = 2, K_z = 3$	7.245E-04	1.781E-04	2.920E-05	4.122E-06	5.466E-07	2.618
$K_y = 3, K_z = 1$	1.361E-02	8.615E-03	4.826E-03	2.547E-03	1.310E-03	0.851
$K_y = 3, K_z = 2$	3.086E-03	8.714E-04	2.279E-04	5.779E-05	1.523E-05	1.924
$K_y = 3, K_z = 3$	4.937E-04	1.423E-04	2.257E-05	3.145E-06	4.168E-07	2.692

$ z_0 - z_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EGH	3.823E-02	1.662E-03	7.548E-03	3.572E-03	1.733E-03	1.122
θ -GH	4.221E-02	1.148E-02	2.897E-03	7.263E-04	1.817E-04	1.964
$K_y = 1, K_z = 1$	1.333E-01	7.399E-02	3.941E-02	1.973E-02	9.976E-03	0.941
$K_y = 1, K_z = 2$	6.109E-02	2.493E-02	7.421E-03	1.952E-03	2.234E-04	2.290
$K_y = 2, K_z = 1$	3.544E-02	1.215E-02	3.446E-02	3.915E-03	2.151E-03	1.005
$K_y = 2, K_z = 2$	5.016E-02	9.158E-03	2.568E-03	9.691E-04	2.204E-04	2.235
$K_y = 2, K_z = 3$	2.360E-03	4.124E-04	5.803E-05	7.725E-06	9.932E-07	2.817
$K_y = 3, K_z = 1$	6.258E-02	3.680E-02	1.963E-03	1.011E-03	5.120E-03	0.908
$K_y = 3, K_z = 2$	1.243E-02	3.142E-03	7.969E-04	2.038E-04	5.100E-05	1.981
$K_y = 3, K_z = 3$	2.314E-03	3.687E-04	4.996E-05	6.428E-06	8.125E-07	2.879

that the multistep scheme (3.7) is still a much more accurate method than the EGH and θ -GH methods. Convergence rates in this example are still similar to those obtained for the one-dimensional BSDE in Example 2.

Example 4. Let us consider the following BSDE:

$$(5.7) \quad -dy_t = \mathbf{A}y_t|y_t|^2 dt - z_t dW_t,$$

where $y_t = \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} 0.5 & -1 \\ 1 & 0.5 \end{pmatrix}$, and $|y_t|^2 = y_1^2(t) + y_2^2(t)$. Note that the generator f in this equation does not depend on z_t . If we set the terminal condition to be

$$y_T = \begin{pmatrix} \sin(W_T + T) \\ \cos(W_T + T) \end{pmatrix},$$

then the analytic solution of (5.7) is given by

$$(5.8) \quad y_t = \begin{pmatrix} \sin(W_t + t) \\ \cos(W_t + t) \end{pmatrix}, \quad z_t = \begin{pmatrix} \cos(W_t + t) \\ -\sin(W_t + t) \end{pmatrix}.$$

TABLE 5.4
Errors and convergence rates for Example 4.

$ y_0 - y_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EGH	1.176E-01	6.053E-02	2.962E-02	1.447E-02	7.115E-03	1.007
θ -GH	6.939E-03	1.464E-03	3.157E-04	9.007E-05	2.715E-05	2.002
$K_y = 1, K_z = 1, 2$	6.939E-03	1.464E-03	3.157E-04	9.007E-05	2.715E-05	2.002
$K_y = 2, K_z = 1, 2, 3$	2.792E-04	2.676E-05	4.110E-06	5.355E-07	5.454E-08	3.029
$K_y = 3, K_z = 1, 2, 3$	8.558E-05	6.111E-06	5.858E-07	2.337E-08	2.817E-09	3.780
$ z_0 - z_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EGH	1.685E-01	9.002E-02	4.660E-02	2.375E-02	1.199E-02	0.955
θ -GH	9.902E-03	2.332E-03	5.604E-04	1.556E-04	4.136E-05	1.971
$K_y = 1, K_z = 1$	8.589E-02	4.304E-02	2.134E-02	1.056E-02	5.246E-03	1.014
$K_y = 1, K_z = 2$	4.904E-03	2.346E-03	7.340E-04	1.999E-04	5.171E-05	1.896
$K_y = 2, K_z = 2$	8.852E-03	2.562E-03	6.833E-04	1.757E-04	4.453E-05	1.914
$K_y = 2, K_z = 3$	1.248E-03	2.264E-04	3.165E-05	2.297E-06	5.407E-07	2.805
$K_y = 3, K_z = 3$	1.175E-03	2.084E-04	2.891E-05	3.745E-06	4.778E-07	2.832

Let us take $T = 1$ again. Then the exact solution at $t = 0$ is $y_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $z_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. We also solve (5.7) numerically by using EGH, θ -GH, and the scheme (3.7) with different step numbers K_y and K_z , respectively. Table 5.4 lists all computation results including convergence rates and errors $|y_0 - y_0^0|$ and $|z_0 - z_0^0|$.

Remark 12. Example 4 is different from Example 1 in that the BSDE (5.7) has two equations, and thus y_t and z_t both have two components. Numerical results listed in Table 5.4 show that convergence rates of the scheme (3.7) for solving such a BSDE with $f = f(t, y_t)$ are still similar to that for Example 1, as discussed in Remark 9.

Example 5. In this example, we consider the one-dimensional evaluation problem in stock markets. Denote by p_t and S_t the bond price and the stock price in a stock market, respectively. Assume that p_t and S_t satisfy

$$(5.9) \quad \begin{cases} dp_t = r_t p_t dt, t \geq 0, \\ p_0 = p, \end{cases}$$

and

$$(5.10) \quad \begin{cases} dS_t = b_t S_t dt + \sigma_t S_t dW_t, t \geq 0, \\ S_0 = x, \end{cases}$$

where r_t is the return rate of the bond, b_t is the expected return rate of the stock, and $\sigma_t > 0$ is the volatility of the stock. We note that r_t , b_t , σ_t , and σ_t^{-1} are all bounded.

An investor with wealth y_t at time t puts π_t money into buying stock and uses $y_t - \pi_t$ to buy bonds. Suppose that the stock pays dividends continuously with a

bounded dividend rate $d(t, S_t)$ at time t . Then the processes y_t and π_t satisfy the following stochastic differential equation [15, 18, 19, 25]:

$$(5.11) \quad -dy_t = -(r_t y_t + (b_t - r_t + d(t, S_t))\pi_t)dt - \sigma_t \pi_t dW_t.$$

Let $z_t = \sigma_t \pi_t$. Then (y_t, z_t) satisfies

$$(5.12) \quad -dy_t = -(r_t y_t + (b_t - r_t + d(t, S_t))\sigma_t^{-1} z_t)dt - z_t dW_t.$$

For the European call option, the terminal condition for (5.12) is given at the mature time T by

$$(5.13) \quad y_T = \max\{S_T - K, 0\},$$

where S_T is the solution S_t of the problem (5.10) at the mature time T and K is the strike price.

When $r_t = r$, $b_t = b$, $\sigma_t = \sigma$, and $d(t, S_t) = d$ are constants in (5.12), we can obtain the analytic solution of (5.12) and (5.13) in the closed form. In this case, the analytic solution of the BSDE (5.12) with the terminal condition (5.13) is given by

$$(5.14) \quad \begin{cases} y_t = V(t, S_t) = S_t e^{-d(T-t)} \mathbf{N}(d_1(S_t)) - K e^{-r(T-t)} \mathbf{N}(d_0(S_t)), \\ z_t = \frac{\partial V}{\partial S} \sigma = S_t e^{-d(T-t)} \mathbf{N}(d_1(S_t)) \sigma, \\ d_0(S_t) = \frac{1}{\sigma \sqrt{T-t}} \ln \left(\frac{S_t}{K e^{(r-d)(T-t)}} \right) - \frac{1}{2} \sigma \sqrt{T-t}, \\ d_1(S_t) = d_0(S_t) + \sigma \sqrt{T-t}, \end{cases}$$

where $\mathbf{N}(\cdot)$ denotes the cumulative normal distribution function. For our test, we set $T = 0.33$, $K = S_0 = 100$, $r = 0.03$, $\mu = 0.05$, $d = 0.04$, and $\sigma = 0.2$ for the BSDE (5.12) with the terminal condition (5.13). Then (5.14) tells us that the exact solution of (5.12) at time $t = 0$ is $(y_0, z_0) \approx (4.3671, 10.0950)$.

We numerically solve (5.12) using EMC, EGH, θ -MC, θ -GH, and the multistep scheme (3.7) with different step numbers K_y and K_z , respectively. Table 5.5 reports all results on convergence rates and errors $|y_0 - y_0^0|$ and $|z_0 - z_0^0|$. Note that N_E in Table 5.5 is the random sampling number used in the methods EMC and θ -MC.

Remark 13. The aim of Example 5 is to show the effectiveness and accuracy of our multistep scheme (3.7) for solving the Black–Scholes equation. Since Monte Carlo methods are very popular in the field of computational finance, we also compare our method with the Monte Carlo methods discussed in [33]. The errors and convergence rates listed in Table 5.5 show that our multistep scheme (3.7) is much more accurate than the implicit Euler methods (EMC and EGH) and the θ -scheme (θ -MC and θ -GH). The EMC and θ -MC methods are the least accurate. It is observed that even the EGH with $N = 2$ and the θ -GH with $N = 8$ can achieve the accuracy of the EMC and θ -MC with $N = 64$ and the sampling number $N_E = 40000$. The θ -GH [33] is more efficient and accurate than the EMC, θ -MC, and EGH methods. However, the accuracy of θ -GH with $N = 64$ can easily be achieved by our multistep method with $N = 16$, $K_y = 1$, or $K_z = 2$, and with $N = 8$, $K_y = 2$, and $K_z = 3$. The results in Table 5.5 again show that convergence rates of the multistep scheme (3.7) for y_t and z_t are both $\min(K_y + 1, K_z)$.

TABLE 5.5
Errors and convergence rates for Example 5.

$ y_0 - y_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EMC	$N_E = 1000$ 8.720E-02	$N_E = 4000$ 4.730E-02	$N_E = 10000$ 3.100E-02	$N_E = 20000$ 2.230E-02	$N_E = 40000$ 1.080E-02	0.650
θ -MC	2.630E-02	6.200E-03	2.000E-03	6.000E-04	2.000E-04	1.791
EGH	1.877E-02	9.646E-03	4.839E-03	2.409E-03	1.205E-03	0.992
θ -GH	1.219E-03	4.893E-04	1.534E-04	3.820E-05	1.251E-05	1.700
$K_y = 1, K_z = 1$	8.429E-04	2.985E-04	5.674E-05	1.128E-05	1.262E-05	1.685
$K_y = 1, K_z = 2$	1.284E-04	2.925E-05	4.236E-06	5.484E-07	8.531E-08	2.685
$K_y = 2, K_z = 2$	6.935E-04	1.888E-04	4.192E-05	1.204E-05	1.357E-06	2.197
$K_y = 2, K_z = 3$	3.742E-06	4.339E-07	2.037E-08	2.738E-10	6.846E-10	3.546
$K_y = 3, K_z = 3$	5.063E-05	1.432E-06	2.583E-07	1.794E-07	3.607E-09	3.055
$ z_0 - z_0^0 $						
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	CR
EMC	$N_E = 1000$ 2.431E-01	$N_E = 4000$ 7.025E-01	$N_E = 10000$ 4.432E-01	$N_E = 20000$ 4.607E-01	$N_E = 40000$ 5.826E-01	0.200
θ -MC	8.730E-01	5.218E-01	2.421E-01	9.890E-02	4.920E-02	0.900
EGH	3.767E-01	1.956E-03	9.980E-02	5.043E-02	2.535E-02	0.974
θ -GH	1.728E-03	3.575E-04	7.667E-05	1.978E-05	1.043E-06	2.629
$K_y = 1, K_z = 1$	9.894E-03	4.965E-03	2.488E-03	1.252E-03	6.288E-04	0.994
$K_y = 1, K_z = 2$	5.204E-04	1.571E-04	4.342E-05	1.132E-05	2.928E-06	1.874
$K_y = 2, K_z = 2$	6.736E-04	1.965E-04	4.317E-05	6.401E-05	5.687E-07	2.536
$K_y = 2, K_z = 3$	4.445E-05	5.523E-06	7.255E-08	2.776E-08	3.791E-09	3.467
$K_y = 3, K_z = 3$	4.101E-05	5.070E-06	9.867E-08	6.277E-08	3.746E-09	3.317

6. Conclusions. In this paper, we propose a stable multistep discretization scheme for solving backward stochastic differential equations (BSDEs) on time-space grids. We make use of the Lagrange interpolating polynomials in time and the Gauss–Hermite quadrature formula in space to approximate conditional mathematical expectations. Polynomial interpolations (such as cubic polynomials) are taken to approximate function values at nongrid points in space. The proposed scheme is uncoupled not only between the processes y_t and z_t , but also among space grids at each time level. It is worth pointing out that the data used in the scheme at each time-space point can be local. When the generator function f of the BSDE is independent of the random variable z_t , we rigorously derive some error estimates for the proposed semidiscrete scheme. We also would like to note that error estimates of the multistep scheme for more general BSDEs are still under our investigation. Various numerical tests and comparisons are presented; the results are consistent with our theoretical results and show that the proposed scheme is highly effective and accurate for solving BSDEs.

Acknowledgment. The authors would like to thank the referees for their valuable suggestions which greatly improved the paper.

REFERENCES

- [1] M. ABRAMOWITZ AND I. STEGUN, *Handbook of Mathematical Functions*, Dover, New York, 1972.
- [2] V. BALLY, *An Approximation Scheme for BSDEs and Applications to Control and Nonlinear PDEs*, Pitman Res. Notes Math. 364, Longman, London, 1997.
- [3] C. BENDER AND R. DENK, *A forward scheme for backward SDEs*, Stochastic Process. Appl., 117 (2007), pp. 1793–1812.
- [4] B. BOUCHARD AND N. TOUZI, *Discrete-time approximation and Monte-Carlo simulation of backward stochastic differential equations*, Stochastic Process. Appl., 111 (2004), pp. 175–206.
- [5] J. BUTCHER, *Numerical Methods for Ordinary Differential Equations*, John Wiley & Sons, New York, 2003.
- [6] D. CHEVANCE, *Numerical methods for backward stochastic differential equations*, in Numerical Methods in Finance, L. C. G. Rogers and D. Talay, eds., Cambridge University Press, London, 1997, pp. 232–244.
- [7] J. CVITANIC AND J. ZHANG, *The steepest descent method for FBSDEs*, Electron. J. Probab., 10 (2005), pp. 1468–1495.
- [8] F. DELARUE AND S. MENOZZI, *A forward-backward stochastic algorithm for quasi-linear PDEs*, Ann. Appl. Probab., 16 (2006), pp. 140–184.
- [9] F. DELARUE AND S. MENOZZI, *An interpolated stochastic algorithm for quasi-linear PDEs*, Math. Comp., 77 (2008), pp. 125–158.
- [10] J. DOUGLAS, JR., J. MA, AND P. PROTTER, *Numerical methods for forward-backward stochastic differential equations*, Ann. Appl. Probab., 6 (1996), pp. 940–968.
- [11] E. R. GIANIN, *Risk measures via g -expectations*, Insurance Math. Econom., 39 (2006), pp. 19–34.
- [12] E. GOBET AND C. LABART, *Error expansion for the discretization of backward stochastic differential equations*, Stochastic Process. Appl., 117 (2007), pp. 803–829.
- [13] E. HAIRER AND G. WANNER, *Solving Ordinary Differential Equations I. Nonstiff Problems*, Springer Ser. Comput. Math. 8, Springer-Verlag, Berlin, 1993.
- [14] E. HAIRER AND G. WANNER, *Solving Ordinary Differential Equations II. Stiff and Differential-Algebraic Problems*, Springer Ser. Comput. Math. 14, Springer-Verlag, Berlin, 1996.
- [15] N. EL KAROUI, S. PENG, AND M. C. QUENEZ, *Backward stochastic differential equations in finance*, Math. Finance, 7 (1997), pp. 1–71.
- [16] O. LADYZENSKAJA, O. SOLONNIKOV, AND N. URALCEVA, *Linear and Quasi-linear Equations of Parabolic Type*, Transl. Math. Monogr. 23, AMS, Providence, RI, 1968.
- [17] J. MA, P. PROTTER, J. SAN MARTIN, AND S. TORRES, *Numerical methods for backward stochastic differential equations*, Ann. Appl. Probab., 12 (2002), pp. 302–316.
- [18] J. MA, P. PROTTER, AND J. YONG, *Solving forward-backward stochastic differential equations explicitly—A four step scheme*, Probab. Theory Related Fields, 98 (1994), pp. 339–359.
- [19] J. MA AND J. YONG, *Forward-Backward Stochastic Differential Equations and Their Applications*, Lecture Notes in Math. 1702, Springer, New York, 1999.
- [20] J. MA AND J. ZHANG, *Representation theorems for backward stochastic differential equations*, Ann. Appl. Probab., 12 (2002), pp. 1390–1418.
- [21] J. MA, J. SHEN, AND Y. ZHAO, *On numerical approximations of forward-backward stochastic differential equations*, SIAM J. Numer. Anal., 46 (2008), pp. 2636–2661.
- [22] J. MÉMIN, S. PENG, AND M. XU, *Convergence of solutions of discrete reflected backward SDE's and simulations*, Acta Math. Appl. Sinica, 24 (2008), pp. 1–18.
- [23] G. N. MILSTEIN AND M. V. TRETYAKOV, *Numerical algorithms for forward-backward stochastic differential equations*, SIAM J. Sci. Comput., 28 (2006), pp. 561–582.
- [24] G. MILSTEIN AND M. TRETYAKOV, *Discretization of forward-backward stochastic differential equations and related quasi-linear parabolic equations*, IMA J. Numer. Anal., 27 (2007), pp. 24–44.
- [25] E. PARDOUX AND S. PENG, *Adapted solution of a backward stochastic differential equation*, Systems Control Lett., 14 (1990), pp. 55–61.
- [26] S. PENG, *A general stochastic maximum principle for optimal control problems*, SIAM J. Control Optim., 28 (1990), pp. 966–979.
- [27] S. PENG, *Probabilistic interpretation for systems of quasilinear parabolic partial differential equations*, Stochastics Stochastics Rep., 37 (1991), pp. 61–74.

- [28] S. PENG, *Backward SDE and related g -expectation*, in Backward Stochastic Differential Equations, Pitman Res. Notes Math. 364, Pitman, Boston, 1997, pp. 141–159.
- [29] S. PENG, *A linear approximation algorithm using BSDE*, Pacific Econom. Rev., 4 (1999), pp. 285–291.
- [30] G. DA PRATO AND J. ZABCYK, *Second Order Partial Differential Equations in Hilbert Spaces*, Cambridge University Press, London, 2002.
- [31] J. ZHANG, *A numerical scheme for BSDEs*, Ann. Appl. Probab., 14 (2004), pp. 459–488.
- [32] Y. ZHANG AND W. ZHENG, *Discretizing a backward stochastic differential equation*, Int. J. Math. Math. Sci., 32 (2002), pp. 103–116.
- [33] W. ZHAO, L. CHEN, AND S. PENG, *A new kind of accurate numerical method for backward stochastic differential equations*, SIAM J. Sci. Comput., 28 (2006), pp. 1563–1581.
- [34] W. ZHAO, J. WANG, AND S. PENG, *Error estimates of the θ -scheme for backward stochastic differential equations*, Discrete Contin. Dyn. Syst. Ser. B, 12 (2009), pp. 905–924.